CRMPro AI

AI-Powered Customer Relationship Management Platform

BUSINESS REQUIREMENTS DOCUMENT

Target market: IT services and software development companies

Document information	Value
Document version	1.0 – Detailed business baseline
Status	Draft for business review and scope confirmation
Prepared for	CRMPro AI product definition and portfolio implementation
Business domain	B2B customer relationship management for IT services
Primary users	Sales, account management, customer support, management and customer portal users
Document scope	Business goals, domain definitions, functional requirements, AI use cases, workflows, rules and acceptance criteria
Out of scope	Technical architecture, database schema, API contracts, UI wireframes and detailed implementation design

Prepared as a product-level business specification

Document Control

Version	Date	Description	Status
1.0	06 November 2026	Initial detailed BRD covering business scope, functional modules, AI requirements and acceptance criteria.	Draft

How to Use This Document

This BRD describes what CRMPro AI must do from a business perspective and why each capability exists. It is intended to be reviewed by a product owner, business analyst, designer, developer and tester before detailed technical design begins.

- Requirements are assigned unique IDs so they can be traced into design, development and testing.
- Priority uses Must, Should and Could. Must requirements define the minimum credible product; Should requirements improve the product after the core workflow works; Could requirements are optional enhancements.
- AI requirements specify the business problem, trigger, information used, expected result, human control and fallback. They do not prescribe a particular model or framework.
- Any assumptions that affect scope are stated explicitly and should be confirmed before implementation.

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1. Executive Summary

CRMPro AI is a multi-organization SaaS customer relationship management platform for small and medium-sized IT services and software development companies. It supports the business relationship from the first enquiry through qualification, sales opportunity, proposal, customer onboarding, ongoing account management and post-sale support.

The product addresses a common operational problem: customer information and commitments are often spread across spreadsheets, email threads, chat applications, calendars and separate support tools. This fragmentation makes it difficult to know who owns a relationship, what was promised, what the next action is, which deals are healthy and whether customers are receiving timely support.

CRMPro AI centralizes the customer record and activity history, makes sales and support work visible, and adds AI assistance where language understanding, retrieval, summarization, prediction or multi-step coordination can reduce effort. AI is an assistant to authorized employees, not an uncontrolled replacement for business judgment.

Product definition
CRMPro AI is a B2B CRM for IT service providers. It is not an ERP, accounting system, software project delivery platform or generic chatbot. Its core business object is the relationship between the IT service provider and a prospect/customer organization.

1.1 Product Vision

Help IT service companies build stronger customer relationships and run a more disciplined sales and support operation by giving every authorized employee a reliable, connected view of prospects, customers, opportunities, interactions, proposals and support issues—with AI assistance grounded in approved business data.

1.2 Business Outcomes

Outcome	How the product supports it	Example measure
Better lead response	Capture enquiries, assign ownership and make follow-ups visible.	Time from lead creation to first meaningful contact.
Improved sales execution	Track deals through clear stages, values, risks and next actions.	Opportunity win rate and stage ageing.
Consistent customer knowledge	Keep contacts, interactions, proposals and support history together.	Percentage of active accounts with complete key information.
More effective support	Organize tickets, ownership, priorities, service targets and reusable knowledge.	First response time, resolution time and SLA attainment.
Less repetitive work	Draft summaries, emails and ticket replies using approved context.	Employee time saved and AI draft acceptance rate.
Better management visibility	Provide consistent dashboards and evidence-based analysis.	Report usage, data completeness and forecast calibration.

1.3 Product Success Principles

- One trusted customer record: users should not need to reconstruct a customer relationship from disconnected screens.
- Clear ownership: every active lead, opportunity and ticket should have an accountable owner or queue.
- Visible next action: active work should have a status, owner and next step where appropriate.
- AI with evidence: AI outputs should show the data or documents on which they rely, and distinguish facts from suggestions.
- Human control: users approve externally visible communications and consequential record changes unless an organization explicitly configures a safe, authorized automation.
- Small-company usability: the system should be powerful without forcing a small IT firm to operate a complex enterprise sales process.

2. Business Background and Problem Statement

2.1 Typical Operating Environment

The target organization sells services such as custom software development, web and mobile application development, cloud and infrastructure services, system integration, IT consulting, application maintenance and technical support. A typical sales journey begins with an enquiry from a website, referral, professional network or existing customer. A sales employee qualifies the requirement, arranges discussions, prepares a proposal and negotiates the engagement. After the deal is won, account and delivery teams continue the relationship. Customers may later request enhancements, report defects or ask for support.

2.2 Current-State Problems

Problem	Operational impact	CRMPro AI response
Scattered prospect records	Duplicate leads, missing contact details and unclear ownership.	Central lead and account records, duplicate review and assignment.
Unstructured follow-up	Potential customers are forgotten or contacted late.	Tasks, due dates, reminders and overdue views.
Invisible sales pipeline	Managers cannot reliably see deal stage, value or risk.	Opportunity stages, history, pipeline reports and risk indicators.
Proposal inconsistency	Different employees use different formats and omit key terms.	Proposal templates, version history and completeness checks.
Lost customer context	Knowledge is held in individual inboxes or employees' memory.	Account timeline, linked activities, documents and customer summaries.
Reactive support	Tickets are missed, duplicated or repeatedly investigated.	Ticket queues, priorities, SLA tracking, knowledge search and similar-ticket discovery.
Manual reporting	Managers spend time collecting and interpreting information.	Consistent dashboards and AI-assisted explanations grounded in CRM data.

2.3 Business Need

The organization needs a single place to manage customer-facing work, while retaining the flexibility of a small or medium-sized IT services business. It also needs practical AI features that reduce repetitive work and help employees find relevant information quickly. The product should not require a large implementation team, complex configuration or extensive data science resources to provide value.

3. Product Definition, Target Market and Boundaries

3.1 Target Organization

The initial target is an IT services company with approximately 5–200 internal users, a B2B sales process, repeat customer relationships and a support or maintenance function. This is a planning assumption for product design, not a technical limit. The application should be usable by a small agency and scalable in process maturity as the organization grows.

3.2 Example Company Scenario

Example: “Northstar Software Solutions” provides custom web applications, mobile applications, cloud migration and application maintenance. A prospect, “ABC Manufacturing Ltd.”, submits an enquiry for a new ERP system. A sales executive qualifies the requirement, creates an opportunity, holds meetings, prepares a proposal and records the outcome. When ABC accepts, it becomes an active customer. Later, ABC raises a support ticket through the customer portal. The support team uses the account history and knowledge base to resolve it. Management uses dashboards and AI-assisted analysis to understand the relationship and identify risks.

3.3 Product Tenancy and Data Separation

The product is intended to serve multiple independent customer organizations (tenants). Each tenant has its own users, customers, leads, deals, proposals, tickets, knowledge articles, settings and AI usage. A tenant must never see another tenant’s records. A platform operator may manage subscription and service-level platform operations but must not browse tenant business content by default.

3.4 Product Boundary

Included in CRMPro AI	Not part of the initial product
Lead, account, contact and opportunity management.	Full accounting, general ledger, tax filing or payroll.
Sales activities, follow-ups and proposal workflow.	Full project delivery management, sprints, source control or time tracking.
Customer support ticketing and customer portal.	A full IT service management suite for internal infrastructure.
Knowledge base and controlled document retrieval.	A general-purpose enterprise document management system.
CRM reports, business assistant and AI-assisted workflows.	Autonomous sales negotiation or unsupervised external communications.
Basic customer onboarding handoff and relationship history.	Full ERP, inventory, procurement or billing platform.

4. Stakeholders, User Groups and Responsibilities

Role / stakeholder	Primary goals	Typical responsibilities
Business owner / executive	Understand revenue opportunities, customer health and operational performance.	Review dashboards, approve key commercial decisions, monitor performance.
Organization administrator	Configure CRM for the company and control access.	Manage users, roles, teams, settings, categories and data policies.
Sales manager	Manage team pipeline and improve conversion.	Assign leads, review opportunities, coach sales executives, approve proposals where required.
Sales executive	Convert enquiries into business.	Qualify leads, maintain contacts, schedule follow-ups, manage opportunities and prepare proposals.

Account manager	Maintain and grow existing customer relationships.	Review account health, coordinate customer reviews, track risks and expansion opportunities.
Support manager	Ensure customer issues are handled effectively.	Manage queues, assignment, escalation, SLA policy and support performance.
Support agent	Resolve customer issues and communicate clearly.	Work assigned tickets, use knowledge articles, record diagnosis and resolution.
Customer portal user	Submit and monitor requests for their own organization.	Create tickets, add comments or attachments, view permitted ticket status and knowledge articles.
Platform operator	Operate the SaaS service across tenants.	Manage platform availability, subscription status and platform-level support without unrestricted tenant-data access.

4.1 Role and Permission Principles

- Permissions must be configurable by role and, where appropriate, by team, record owner and organization.
- Sales executives may be limited to their own leads and opportunities, while managers may see their team or organization pipeline.
- Support agents should see the customer and ticket information required to resolve assigned work, not unrelated commercial data by default.
- Customer portal users may see only their own organization's permitted contacts, tickets, responses and shared documents.
- AI assistants and agents must use the same data-access rules as the requesting user. AI must not bypass permissions.
- Sensitive actions such as exporting large datasets, deleting records, changing ownership or sending external communications require explicit permissions and audit history.

4.2 Suggested Role Matrix

Capability	Org Admin	Sales Manager	Sales Exec	Account Manager	Support Manager	Support Agent	Customer
Manage users / settings	Yes	No	No	No	No	No	No
View full sales pipeline	Yes	Team / org	Own / permitted	Assigned accounts	Limited	No by default	No
Create and update leads	Yes	Yes	Yes	Limited	No	No	No
Manage accounts / contacts	Yes	Yes	Yes	Yes	Read limited	Read limited	Own org limited
Manage proposals	Yes	Approve / view	Create / edit	View permitted	No	No	View shared only
Manage support tickets	Yes	Limited	Limited	View assigned account	Yes	Assigned / queue	Own org tickets
Manage knowledge articles	Yes	Approve / manage	Read	Read	Approve / manage	Contribute by permission	Published only
Use AI	Yes	Yes	Yes	Yes	Yes	Yes	Limited

assistant							future option
Approve AI-generated actions	By permission	By permission	Own authorized drafts	By permission	By permission	By permission	No

The matrix is a starting point. The organization administrator should be able to assign granular permissions, but the product must provide sensible role templates so setup is not burdensome.

5. CRM Domain Concepts and Business Vocabulary

Concept	Business meaning	Key distinction / relationship
Lead	An unqualified or not-yet-converted expression of interest from a person or organization.	A lead is a prospect record, not automatically a customer or a deal.
Account	An organization with which the company may have a prospect or customer relationship.	An account can exist before a sale. Its relationship status indicates prospect, customer or former customer.
Contact	A person associated with an account or lead.	One account can have many contacts; a contact may be a decision-maker, technical evaluator or billing contact.
Opportunity / Deal	A specific potential sale with a defined need, estimated value and expected decision.	One account can have multiple opportunities; an opportunity moves through a sales process.
Proposal / Quotation	A commercial response to a customer opportunity.	A proposal can have multiple revisions and must retain its approval and sending history.
Activity	A task or interaction such as a call, email, meeting, demo or follow-up.	Activities should link to the relevant lead, account, contact, opportunity or ticket.
Ticket	A customer-reported issue, question, request or service incident requiring a support response.	A ticket has an owner, priority, status, communication history and resolution.
Knowledge article	An approved reusable answer, instruction, FAQ or policy.	Published articles can be used in RAG; drafts and restricted content must not be exposed.
Customer portal user	A person from a customer organization who accesses a limited external portal.	Portal access is scoped to their organization and permitted records.
Tenant	An independent company using CRMPro AI.	Tenant data and configuration must be logically separated from every other tenant.

5.1 Important Domain Distinctions

- Lead versus Account: a lead is an initial prospect record. An account is the organization record used to consolidate people and business relationships. An account can be marked Prospect before becoming a Customer.
- Account versus Contact: the account represents the company; the contact represents a person at that company.
- Account versus Opportunity: an account is the ongoing relationship; an opportunity is one possible sale. A customer may have several opportunities over time.
- Opportunity versus Proposal: an opportunity tracks the deal; a proposal is a versioned commercial document prepared as part of that deal.

- Ticket versus Opportunity: a ticket is post-sale support or service communication; an opportunity is a potential commercial transaction. A feature request may create a future opportunity, but the records remain distinct.
- CRM versus Project Management: CRM tracks the customer and commercial/support relationship. Delivery plans, sprints and developer tasks belong in a project management system and may be linked later.

5.2 Customer Relationship Lifecycle

Stage	Typical event	System outcome
Awareness / enquiry	A person submits a website form, referral or direct enquiry.	Lead is captured and source is recorded.
Qualification	Sales checks fit, need, budget, authority and timing.	Lead is qualified, disqualified or nurtured; next action is recorded.
Opportunity	A specific service requirement and buying process are identified.	Opportunity is created and assigned to an owner.
Proposal / negotiation	Scope, commercial terms and timing are discussed.	Proposal versions and negotiation activities are recorded.
Won / customer activation	The customer accepts the proposal or confirms the engagement.	Opportunity is marked Won; account relationship is set to Customer; handoff is initiated.
Account management	The company maintains the relationship and identifies needs.	Account reviews, communications and new opportunities are recorded.
Support / maintenance	Customer reports an issue or requests assistance.	Ticket is created, assigned, resolved and retained in account history.
Renewal / expansion	Customer needs ongoing support, additional work or a renewal.	A renewal or expansion opportunity is created and tracked.

6. Scope, Release Boundaries and Prioritization

6.1 In-Scope Capabilities

- Organization setup, users, roles, teams and tenant-level configuration.
- Lead capture, qualification, assignment, follow-up, duplicate handling and conversion.
- Account and contact management with consolidated customer history.
- Opportunity pipeline, stage history, sales activities, value and expected close date.
- Proposal/quotation preparation, templates, versions and approval states.
- Activity, task, meeting and communication timeline.
- Customer support ticket lifecycle, assignment, SLA tracking and customer portal.
- Knowledge article and document management for approved company content.
- Operational dashboards and standard sales, customer and support reports.
- AI-assisted summaries, content drafts, knowledge retrieval, ticket triage and controlled business workflows.
- Audit history, notifications, import/export and basic integrations needed for useful operation.

6.2 Out of Scope for the Initial Product

- Accounting ledger, tax calculation/filing, payroll, payment collection and full invoicing.
- Project delivery management such as sprint planning, developer time tracking, source-code management or deployment management.

- Full marketing automation, mass campaign management and advertising attribution.
- Contact-center telephony, call recording infrastructure and social media publishing.
- Contract e-signature and legally binding automated contract generation.
- Autonomous AI negotiation, autonomous pricing, automatic contract acceptance or unsupervised customer emails.
- Industry-specific clinical, banking or insurance decision functions.
- Large-scale ERP, procurement, inventory, manufacturing or field-service operations.

6.3 Priority Definitions

Priority	Meaning	Release expectation
Must	Essential for a usable, safe business workflow or required to meet the product purpose.	Required for the first credible live release unless explicitly marked phase-later.
Should	Important improvement that increases usability, efficiency or management insight.	Target after the core workflow is stable; may be included when capacity permits.
Could	Useful enhancement that is not required to prove the core product.	Optional future release.

7. Functional Business Requirements by Module

Each requirement below describes an observable business capability. Priority is a product-planning indication. Acceptance criteria appear in Section 15 and should be expanded into test cases during delivery.

7.1 Organization Setup and Tenant Administration

Purpose: Create an independent workspace for each subscribing IT company and maintain its organization profile, settings and business configuration.

Primary users: Platform operator and organization administrator.

AI involvement: AI is not needed for routine tenant setup. AI configuration and usage limits are governed here.

ID	Business requirement	Priority
ORG-001	The platform shall allow an organization to register or be provisioned with a unique tenant workspace.	Must
ORG-002	The organization administrator shall maintain company name, logo, address, website, industry, time zone, default currency and business contact details.	Must
ORG-003	The organization shall configure its sales stages, lead statuses, lead sources, ticket categories and priorities using managed values.	Must
ORG-004	The organization shall configure its default business hours, holidays and support response targets.	Should
ORG-005	The organization shall configure notification preferences and permitted email sender details.	Should
ORG-006	The organization shall manage	Must

	permitted AI features, usage limits, document visibility rules and data-retention preferences.	
ORG-007	The platform shall prevent one tenant from viewing, searching, exporting or retrieving another tenant's data.	Must
ORG-008	The platform shall allow an organization to deactivate its workspace through a controlled process and define data export/retention handling.	Should

7.2 User, Team, Role and Permission Management

Purpose: Give each employee an individual identity and access only to records and actions required for their job.

Primary users: Organization administrator, managers, employees and platform operator.

AI involvement: AI must inherit the current user's permissions; it must not create a separate access path.

ID	Business requirement	Priority
USR-001	An administrator shall invite, activate, deactivate and update internal users.	Must
USR-002	The system shall provide default role templates for administrator, sales manager, sales executive, account manager, support manager and support agent.	Must
USR-003	An administrator shall assign users to one or more teams and define reporting relationships.	Should
USR-004	Permissions shall control access to modules, records, sensitive fields, exports and business actions.	Must
USR-005	Managers shall be able to view records belonging to their permitted team or organization scope.	Must
USR-006	The system shall record significant user and administrator actions in an audit history.	Must
USR-007	A customer portal user shall have a separate limited access profile linked to one customer account.	Must
USR-008	The system shall support deactivation without deleting historical ownership and activity records.	Must

7.3 Dashboard and Work Queue

Purpose: Give each role a useful starting view showing key performance indicators and work requiring attention.

Primary users: Business owner, sales manager, sales executive, account manager, support manager and support agent.

AI involvement: AI can explain patterns and highlight exceptions; KPI totals and charts must be calculated from CRM records.

ID	Business requirement	Priority
DAS-001	The dashboard shall display role-appropriate KPIs for leads, opportunities, customers, activities and support tickets.	Must
DAS-002	Users shall be able to filter dashboard information by date range, owner, team and permitted business dimensions.	Must
DAS-003	The dashboard shall show tasks due today, overdue follow-ups, upcoming meetings and assigned tickets.	Must
DAS-004	The sales dashboard shall show opportunity counts and values by stage, expected close date and owner.	Must
DAS-005	The support dashboard shall show open tickets by priority, status, assignee and SLA risk.	Must
DAS-006	The dashboard shall provide drill-down from a KPI or chart to the underlying permitted records.	Should
DAS-007	The system shall display AI-generated insights separately from verified KPI values and identify the period and data basis.	Should
DAS-008	Users shall be able to dismiss or mark an AI insight as useful/not useful, with feedback recorded for evaluation.	Could

7.4 Lead Management

Purpose: Capture, qualify, assign and convert potential business enquiries into accounts, contacts and opportunities.

Primary users: Sales executive, sales manager and administrator.

AI involvement: AI can summarize free text, suggest qualification questions and draft follow-ups. Predictive lead scoring is a later capability dependent on sufficient outcome data.

ID	Business requirement	Priority
LED-001	Users shall create leads manually with person, organization, contact details, enquiry summary, source and owner.	Must
LED-002	The system shall capture leads from a public enquiry form or approved import channel.	Should
LED-003	Users shall classify leads using configurable status, source, industry, service interest and priority values.	Must
LED-004	Users shall assign or reassign leads to authorized employees or teams and retain assignment history.	Must
LED-005	Users shall record qualification information including need, budget range, decision process, authority/contact role and expected timeline where known.	Must
LED-006	The system shall detect likely duplicate leads using configurable matching	Must

	signals and allow authorized users to review them.	
LED-007	Users shall add notes, calls, meetings, emails, tasks and follow-up dates to a lead timeline.	Must
LED-008	The system shall identify leads with no owner, no next action or overdue follow-up.	Must
LED-009	Authorized users shall convert a qualified lead into an account, contact and one or more opportunities without losing the original lead history.	Must
LED-010	The system shall record a reason when a lead is disqualified or marked lost/nurture.	Should
LED-011	Users shall import leads from CSV with field mapping, validation, duplicate review and an import result summary.	Should
LED-012	AI shall generate an editable lead summary and suggest missing qualification questions from provided lead information.	Should
LED-013	AI shall generate a follow-up email draft using the lead record and approved interaction history.	Should
LED-014	A future predictive score may rank leads using historical conversion outcomes and explain the main contributing factors.	Could

7.5 Account and Contact Management

Purpose: Maintain the organization-level relationship and the people who participate in the buying, delivery and support relationship.

Primary users: Sales, account management, support and administrator.

AI involvement: AI can assemble a Customer 360 summary from records the user is permitted to see; factual account fields remain system-managed.

ID	Business requirement	Priority
ACC-001	Users shall create and maintain an account with organization name, website, industry, address, relationship status and owner.	Must
ACC-002	An account shall have a relationship status such as Prospect, Customer, Inactive Customer or Former Customer.	Must
ACC-003	Users shall associate multiple contacts with an account and identify primary, technical, commercial and billing contacts where relevant.	Must
ACC-004	The system shall maintain account-level links to leads, opportunities, proposals, activities, tickets and shared documents.	Must
ACC-005	Users shall record account notes, relationship review dates, service interests and important business	Should

	context.	
ACC-006	The system shall show a chronological account timeline of permitted customer-facing interactions and business events.	Must
ACC-007	Users shall merge duplicate accounts or contacts through a controlled review process that preserves links and audit history.	Should
ACC-008	AI shall generate a Customer 360 summary with references to the underlying records and clearly identify the date range covered.	Should
ACC-009	AI may highlight account engagement or support-risk signals using defined evidence such as inactivity or unresolved high-priority tickets.	Could

7.6 Opportunity and Sales Pipeline Management

Purpose: Track each potential commercial engagement from discovery through proposal, negotiation and final outcome.

Primary users: Sales executive, sales manager, account manager and administrator.

AI involvement: AI can summarize opportunity context and flag evidence-based risks. ML deal scoring should be enabled only after reliable historical outcomes exist.

ID	Business requirement	Priority
OPP-001	Users shall create an opportunity linked to an account and, where known, a primary contact.	Must
OPP-002	Each opportunity shall include name, service type, description, owner, estimated value, currency, expected close date and stage.	Must
OPP-003	The organization shall configure its sales pipeline stages and permitted stage transitions.	Must
OPP-004	The system shall record each stage change, timestamp, user and optional reason.	Must
OPP-005	Users shall record qualification details, customer needs, competitors, decision criteria and next steps.	Should
OPP-006	The system shall show pipeline value by stage, owner and expected closing period.	Must
OPP-007	An opportunity marked Won shall require a recorded outcome date and final value.	Must
OPP-008	An opportunity marked Lost shall require a loss reason selected from configurable values, with optional notes.	Must
OPP-009	The system shall identify stale opportunities based on configurable inactivity thresholds.	Should
OPP-010	AI shall summarize opportunity history and recommend a next action with	Should

	supporting evidence.	
OPP-011	AI may identify risk signals such as inactivity, repeated date slippage or missing decision-maker engagement.	Should
OPP-012	A predictive win-probability model may be introduced after sufficient representative historical data is available and validated.	Could

7.7 Activity, Task, Meeting and Communication Management

Purpose: Make customer-facing work visible and preserve the interaction history for leads, accounts and opportunities.

Primary users: All internal business users.

AI involvement: AI can summarize notes/transcripts and draft messages. Scheduling, ownership and due-date rules remain deterministic.

ID	Business requirement	Priority
ACT-001	Users shall create tasks linked to a lead, account, contact, opportunity or ticket.	Must
ACT-002	Each task shall have an owner, due date, status and priority.	Must
ACT-003	Users shall record calls, meetings, demonstrations and other interactions with date, participants, notes and outcome.	Must
ACT-004	The system shall provide upcoming, due-today and overdue activity views.	Must
ACT-005	Users shall schedule meetings and record decisions and action items.	Should
ACT-006	The system shall retain a communication timeline and associate messages with relevant CRM records.	Should
ACT-007	AI shall convert user-provided meeting notes or permitted transcripts into a summary, decisions and action items.	Should
ACT-008	AI shall draft a follow-up message using the approved interaction context; the user must review it before sending.	Should
ACT-009	The system shall send reminders for assigned upcoming and overdue activities according to user preferences.	Must

7.8 Proposal and Quotation Management

Purpose: Create consistent commercial proposals connected to an opportunity and retain revisions, approvals and customer response history.

Primary users: Sales executive, sales manager, account manager and administrator.

AI involvement: AI may prepare editable drafts and check completeness. Pricing, discounts, contractual terms and approval remain controlled by authorized people.

ID	Business requirement	Priority
PRO-001	Users shall create a proposal linked to	Must

	an opportunity and customer account.	
PRO-002	A proposal shall contain a title, scope summary, deliverables, assumptions, exclusions, timeline, commercial amount, payment terms and validity period as applicable.	Must
PRO-003	The system shall support reusable templates and organization-approved standard wording.	Should
PRO-004	The system shall maintain version history and identify the current approved/sent version.	Must
PRO-005	The proposal shall follow configurable statuses such as Draft, In Review, Approved, Sent, Accepted, Rejected and Expired.	Must
PRO-006	The system shall support an internal approval step before a proposal is sent when required by policy.	Should
PRO-007	Users shall record when a proposal is sent and the customer's response.	Must
PRO-008	AI shall produce an editable first draft using opportunity requirements and approved templates.	Should
PRO-009	AI shall check for missing sections or inconsistencies and explain each finding.	Should
PRO-010	AI shall not invent prices, promise delivery dates or alter approved legal/commercial wording without explicit user input.	Must

7.9 Customer Support and Ticket Management

Purpose: Capture, prioritize, assign, track and resolve post-sale customer issues and requests.

Primary users: Customer portal user, support agent, support manager, account manager and administrator.

AI involvement: AI can classify tickets, summarize long threads, retrieve approved solutions with RAG and draft replies. Humans remain responsible for technical diagnosis and customer communication.

ID	Business requirement	Priority
TKT-001	Internal users and permitted customer portal users shall create tickets with subject, description, category and relevant attachments.	Must
TKT-002	Each ticket shall have a unique ticket reference and be linked to a customer account and, where available, a contact.	Must
TKT-003	Tickets shall support configurable priority and status values.	Must
TKT-004	Tickets shall be assigned to a support queue or named agent and retain assignment history.	Must
TKT-005	The system shall maintain public replies separately from internal notes.	Must
TKT-006	The system shall track first response and resolution targets according to	Should

	business hours, priority and configured SLA policy.	
TKT-007	The system shall warn users and managers when a ticket is approaching or breaching an SLA target.	Should
TKT-008	The system shall support escalation, reassignment, reopening and resolution codes.	Must
TKT-009	The customer portal user shall be able to view permitted tickets for their organization and add replies or attachments.	Must
TKT-010	The system shall identify possible duplicate or related tickets for employee review.	Should
TKT-011	AI shall suggest ticket category and priority from the description and available context, while allowing the user to override it.	Should
TKT-012	AI shall summarize ticket conversations and highlight the reported issue, steps tried and outstanding questions.	Should
TKT-013	AI shall retrieve relevant approved knowledge articles and similar resolved tickets for support staff.	Should
TKT-014	AI shall draft a proposed customer response with cited sources when knowledge retrieval is used.	Should
TKT-015	AI shall state when no reliable knowledge source is found rather than inventing a technical fix.	Must

7.10 Knowledge Base and Document Management

Purpose: Create and govern reusable company knowledge that employees can search and that AI can use as a trusted retrieval source.

Primary users: Knowledge administrator, support manager, support agents, sales users and permitted internal users.

AI involvement: This module is the controlled content foundation for RAG; document permission and publication status must be respected at retrieval time.

ID	Business requirement	Priority
KB-001	Authorized users shall create knowledge articles with title, category, content, tags, owner and status.	Must
KB-002	The system shall support uploading common business document formats such as PDF, DOCX and TXT, subject to defined file limits.	Should
KB-003	Documents and articles shall support Draft, In Review, Published, Archived and Superseded states.	Must
KB-004	The system shall maintain article/document version history and ownership.	Should
KB-005	The system shall support access	Must

	classification such as organization-wide, team-restricted and customer-specific.	
KB-006	Users shall search published articles using keyword search and filters.	Must
KB-007	The system shall identify source title and relevant section/page when an AI answer uses a document.	Must
KB-008	The system shall exclude drafts, archived or unauthorized content from general AI retrieval.	Must
KB-009	Authorized users shall review and publish AI-assisted knowledge article drafts before they become trusted content.	Should

7.11 AI Business Assistant and Agent Workflows

Purpose: Allow employees to ask questions in natural language and initiate controlled multi-step CRM workflows without navigating every screen.

Primary users: Authorized internal employees; customer-facing AI is not in the initial scope.

AI involvement: This is the main GenAI experience combining structured-data tools, RAG, LLM reasoning and approval-based agent actions.

ID	Business requirement	Priority
AI-001	The assistant shall understand natural-language questions about CRM records within the requesting user's access permissions.	Must
AI-002	For structured questions, the assistant shall retrieve data using approved CRM business operations and present the relevant records or calculations.	Must
AI-003	For questions about policies, manuals and internal documents, the assistant shall retrieve permitted published knowledge and cite sources.	Must
AI-004	The assistant shall support multi-turn conversations and use prior context only within the current user's authorized scope.	Should
AI-005	The assistant shall distinguish verified record values, calculated metrics, model estimates and generated suggestions.	Must
AI-006	The assistant shall provide a clear "insufficient information" response when it cannot support an answer.	Must
AI-007	The assistant shall support predefined workflows such as customer summary, stalled opportunity review and ticket analysis.	Should
AI-008	Agent workflows shall use an explicit set of approved business actions and shall not execute arbitrary instructions or unrestricted queries.	Must
AI-009	Actions that send external messages, materially change records or affect	Must

	customer commitments shall require approval from an authorized user.	
AI-010	The system shall record the initiating user, workflow, tools invoked, records affected, approvals and outcome for agent actions.	Must
AI-011	Users shall be able to provide feedback on an AI answer or recommendation.	Should
AI-012	The organization shall be able to configure AI usage limits and disable selected AI capabilities.	Must

7.12 Reporting and Analytics

Purpose: Provide consistent operational and management reporting from CRM records.

Primary users: Business owner, managers and authorized employees.

AI involvement: LLMs can explain report findings, but totals, ratios and time comparisons must be calculated from defined business data and metric definitions.

ID	Business requirement	Priority
RPT-001	The system shall provide standard lead, sales, account, activity and support reports.	Must
RPT-002	Reports shall support date, owner, team, stage, source and other relevant filters.	Must
RPT-003	Users shall drill down from report summaries to the permitted underlying records.	Should
RPT-004	The system shall define consistent calculation rules for conversion, win rate, pipeline value, response time and resolution time.	Must
RPT-005	Authorized users shall export permitted reports to common formats such as CSV.	Should
RPT-006	AI shall generate a narrative explanation of selected reports using the calculated results and identify the reporting period.	Should
RPT-007	The system shall identify data gaps that make a metric unreliable, such as missing lead source or outcome reasons.	Should

7.13 Notifications and Reminders

Purpose: Bring important work to the attention of the right person at the right time.

Primary users: All internal users and customer portal users for permitted ticket updates.

AI involvement: AI is not required for basic reminders. AI may prioritize or summarize notifications later, but triggers and delivery rules remain deterministic.

ID	Business requirement	Priority
NTF-001	The system shall notify users when a lead, opportunity or ticket is assigned to them.	Must
NTF-002	The system shall remind users about	Must

	due and overdue activities according to preferences.	
NTF-003	The system shall notify relevant users of ticket replies, escalation and SLA risk.	Must
NTF-004	The system shall notify users when a proposal requires review or approval.	Should
NTF-005	Users shall be able to configure permitted notification channels and frequency.	Should
NTF-006	The system shall avoid duplicate or excessive reminders for the same event.	Should

7.14 Search, Import, Export and Audit

Purpose: Help users find records, move initial data into the CRM and trace important changes.

Primary users: All authorized internal users; administrators for bulk operations.

AI involvement: Semantic search can improve discovery, but exact identifiers and structured filters must remain available.

ID	Business requirement	Priority
DAT-001	Users shall search leads, accounts, contacts, opportunities, activities and tickets using permitted fields.	Must
DAT-002	The system shall support filters and sorting for large lists.	Must
DAT-003	Administrators shall import supported records from CSV using field mapping and validation.	Should
DAT-004	Import processing shall produce a row-level result for accepted, rejected and duplicate records.	Must
DAT-005	Bulk export shall require permission and respect record-level access restrictions.	Must
DAT-006	The system shall preserve audit details for critical record creation, edits, ownership changes, status transitions, export and deletion.	Must
DAT-007	Users shall be able to search knowledge content semantically where that capability is enabled.	Should
DAT-008	AI document retrieval shall apply tenant, role, team and document-level access filters before content is provided to the model.	Must

8. End-to-End Business Processes

8.1 Process A – Enquiry to Qualified Lead

Step	Business action	Responsible role	System behaviour / control
1	A prospect submits an enquiry or an employee	Prospect / sales executive	Create lead with source, enquiry details and

	records a referral.		timestamp.
2	The system checks for possible duplicates.	System; sales employee reviews	Show possible matches; do not silently merge uncertain records.
3	A manager or routing rule assigns ownership.	Sales manager	Set owner and retain assignment history.
4	Sales reviews the requirement and contacts the prospect.	Sales executive	Record call/email/meeting and next action.
5	AI may summarize the enquiry and suggest qualification questions.	Sales executive	Display editable suggestions with the original input available.
6	Sales records qualification outcome.	Sales executive	Mark qualified, nurture or disqualified; record reason where appropriate.
7	A qualified requirement becomes an opportunity.	Sales executive	Convert or create linked account/contact/opportunity without losing history.

8.2 Process B – Opportunity to Won Customer

Step	Business action	Responsible role	System behaviour / control
1	Create an opportunity for a defined service requirement.	Sales executive	Link account/contact, value, stage, owner and expected close date.
2	Hold discovery and technical discussions.	Sales / technical participant	Record requirements, decision criteria, competitors and activities.
3	Prepare a proposal.	Sales executive	Use approved template, version and internal review where configured.
4	AI may draft or check the proposal.	Sales executive / approver	AI output remains draft; commercial values come from user-approved inputs.
5	Send proposal and record customer response.	Authorized sales user	Record sent version, date and outcome.
6	Negotiate and update opportunity stage.	Sales executive / manager	Maintain stage history, value changes and next action.
7	Mark Won or Lost.	Authorized sales user	Require final value/date for Won and loss reason for Lost.
8	If Won, initiate customer handoff.	Sales / account manager	Update account status and create onboarding actions.

8.3 Process C – Customer Onboarding and Account Handoff

When an opportunity is won, the CRM should make the transition from sales to account management visible. The handoff should capture the customer's stated objectives, purchased service, proposal version, agreed scope summary, key contacts, known risks, promised milestones and open actions. CRMPro AI does not manage delivery execution, but it should preserve the commercial context so delivery and account teams do not need to reconstruct it.

- The account relationship status becomes Customer only after the organization's defined conversion condition is met (for example, signed acceptance or authorized confirmation).

- The sales owner remains in the history; an account manager may become the ongoing relationship owner.
- Handoff completeness should be checked before the opportunity is considered fully transitioned.

8.4 Process D – Customer Ticket to Resolution

Step	Business action	Responsible role	System behaviour / control
1	Customer or employee submits an issue.	Customer / employee	Create ticket, link account/contact and acknowledge receipt.
2	Ticket is classified and prioritized.	Support team; AI may suggest	Store final category/priority and allow override.
3	Ticket is assigned.	Support manager / routing	Set queue/agent and response target.
4	Support reviews history and knowledge.	Support agent	Show account context, related tickets and permitted knowledge articles.
5	AI retrieves relevant guidance and drafts a response.	Support agent	Cite source article or similar ticket; label as suggestion.
6	Agent investigates and communicates.	Support agent	Separate internal notes from customer-visible replies.
7	Ticket is resolved and customer is informed.	Support agent	Record resolution code, summary and timestamps.
8	Customer confirms or ticket is closed per policy.	Customer / support agent	Allow reopening within configured rules and retain full history.

8.5 Process E – Customer 360 Review

1. An authorized user opens an account and selects “Generate customer summary”.

2. The system retrieves only permitted account, contact, opportunity, activity, proposal and ticket records.

3. The assistant organizes the information into relationship overview, current sales, open issues, recent interactions and suggested next actions.

4. Every factual statement should link to or identify its supporting CRM record where practical.

5. The user reviews the summary and may save it as an internal note or export it according to permission.

9. AI Strategy and Detailed AI Requirements

9.1 AI Design Principles

Principle	Business requirement
Use AI only when useful	Do not use an LLM for deterministic calculations, ordinary CRUD, mandatory field validation or simple workflow transitions.
Ground outputs	Use authorized CRM records for structured facts and published knowledge for document-based answers.
Preserve access control	The user’s permissions and tenant boundary apply before retrieval, during tool use and when presenting results.
Show evidence	Provide record references, article titles, document sections or a clear explanation of the data basis.
Human approval	Draft external messages and consequential updates; require

	explicit approval before sending or committing.
Explain uncertainty	Distinguish known facts from model estimates and suggestions. Abstain when evidence is missing.
Measure quality	Evaluate retrieval relevance, groundedness, classification quality, task success, user acceptance and cost.
Protect data	Do not use tenant content to train shared models without explicit, informed agreement and appropriate safeguards.

9.2 AI Capability Types Used in the Product

Capability	Meaning in this product	Typical use
LLM generation	Generate or transform natural language using supplied context.	Email drafts, summaries, proposal drafts, report narratives.
RAG	Retrieve relevant approved content and provide it to an LLM to answer with evidence.	Knowledge articles, SOPs, product manuals, troubleshooting documents.
Tool calling	Allow an LLM-driven workflow to request approved CRM operations.	Look up a lead, list open tickets, create a draft task, prepare a report.
Agent orchestration	Coordinate a bounded multi-step workflow using tools and possibly specialized task roles.	Analyze stalled deals, summarize an account, prepare ticket analysis.
Machine learning	Use historical labeled business outcomes to estimate a future outcome.	Lead conversion propensity, opportunity win probability; only after data validation.
Classification / extraction	Assign a category or extract structured fields from text/documents.	Ticket category suggestion, lead service-interest extraction, document metadata.
Semantic similarity	Find records or documents with similar meaning rather than exact wording.	Similar resolved tickets and knowledge article discovery.

9.3 AI Feature Catalogue

AI feature	Where used	Inputs and trigger	Output and business value	Human control / fallback
Lead intake summary	Lead Management	New lead or employee action; enquiry text and permitted lead fields.	Concise need, organization context, service interest and missing information.	Sales user edits; original source remains visible.
Qualification coach	Lead Management	Sales user asks for next questions.	Suggested questions about need, budget, authority, timing and fit.	Suggestions only; sales employee decides qualification.
Lead scoring (future ML)	Lead Management	Scoring request or scheduled scoring; historical lead attributes and outcomes.	Estimated conversion propensity and key contributing factors.	Do not enable until data sufficiency and validation gate pass; show score as estimate.
Follow-up draft	Lead / Activity	User requests a draft after a call, email or inactivity.	Personalized editable email with relevant context.	User reviews and sends; AI cannot invent commitments.
Customer 360 summary	Account	User requests an account overview.	Summary of relationship, deals, activities and support issues.	All claims link to records where possible; permission-filtered retrieval.
Deal risk assistant	Opportunity	Opportunity review or inactivity signal.	Evidence-based risks and recommended next step.	Do not treat generated risk as fact; show specific signals.

Win probability (future ML)	Opportunity	Opportunity scoring; historical won/lost cases and current features.	Calibrated probability estimate.	Requires representative data, leakage checks, bias review and monitoring.
Proposal drafting	Proposal	User starts proposal from an opportunity.	Editable scope and narrative based on approved input/template.	Price, dates, legal terms and scope require human confirmation.
Proposal completeness review	Proposal	User requests review before approval.	Missing sections, inconsistent details and questions to resolve.	Advisory only; approver owns final version.
Meeting summary	Activity	User supplies notes or an authorized transcript.	Summary, decisions, action items and owners.	User confirms before saving or creating tasks.
Ticket classification	Support Ticket	Ticket creation or update.	Suggested category, issue type and priority with confidence.	Agent can override; low confidence routes to manual triage.
Ticket conversation summary	Support Ticket	Long thread or new agent assignment.	Reported issue, timeline, tests tried, current state and missing facts.	Agent checks source thread; no fabricated troubleshooting.
RAG support answer	Ticket / Knowledge	Agent asks a question or requests guidance.	Answer grounded in published knowledge with source references.	No source = say so; agent validates before customer response.
Similar ticket discovery	Ticket / Knowledge	Ticket is opened or agent searches.	Previously resolved similar cases and their resolutions.	Similarity is not proof of same cause; agent reviews applicability.
AI reply draft	Ticket	Agent requests a response.	Customer-friendly reply grounded in ticket and permitted knowledge.	Agent approves and sends; no autonomous external reply in initial release.
Natural-language CRM analyst	AI Assistant	User asks a structured business question.	Filtered records, calculations and explanation.	Metrics calculated deterministically; query scope and data period shown.
Agentic report preparation	AI Assistant	Manager requests a defined analysis.	Multi-step retrieval, grouping and narrative report.	Read-only initial workflow; user reviews report before sharing.
Agentic follow-up preparation	AI Assistant	Manager requests follow-up actions for a filtered set.	Draft tasks or message drafts with proposed owners and dates.	Bulk actions require preview and explicit approval.

9.4 Where RAG Is Required

RAG is primarily for information that is stored in documents or knowledge articles and cannot be answered reliably from a simple structured record lookup. The initial knowledge sources should be:

- Published support FAQs and troubleshooting articles.
- Product and service documentation maintained by the IT company.
- Approved internal SOPs for ticket handling, escalation and customer communication.
- Approved sales playbooks, proposal guidance and service descriptions.
- Customer-specific documents only when the user has permission to access that customer and the document is explicitly classified for that audience.

RAG should not be the primary way to answer “How many open tickets are there?” or “What is the total value of open deals?” Those answers must be based on structured CRM records and deterministic aggregation. The document assistant should cite the source document and relevant section/page where available.

9.5 Where AI Agents Are Required

The first release should use bounded agent workflows rather than open-ended autonomous agents. A workflow may coordinate specialized responsibilities (for example, sales analysis and support analysis), but it should use a controlled set of tools and a clear objective. “Multi-agent” is a business orchestration pattern, not a requirement to create many independent autonomous bots.

Agent / workflow	Business objective	Permitted tools	Initial autonomy
Lead Assistant	Help qualify and follow up on new enquiries.	Read lead, read account match, create draft note, draft email, propose task.	Suggest and prepare drafts; employee confirms.
Sales Pipeline Analyst	Find stalled, high-value or at-risk opportunities.	List/filter opportunities, read activities, calculate defined metrics, draft tasks.	Read-only analysis and draft follow-ups.
Support Triage Assistant	Prepare new tickets for efficient human handling.	Read ticket, suggest category/priority, retrieve knowledge, find similar tickets.	May save suggested classification after agent review.
Knowledge Assistant	Answer questions using approved internal documents.	Search published articles/documents under permission filters.	Read-only; no source means abstain.
Customer 360 Workflow	Combine account, sales and support context.	Read account, contacts, opportunities, activities, tickets and approved docs.	Read-only summary.
Reporting Workflow	Prepare a requested business report.	Run approved report queries, calculate metrics and summarize results.	Read-only; sharing/export requires user action.

9.6 AI Workflow Example – “Prepare My Weekly Sales Review”

1. Manager asks: “Prepare my weekly sales review for my team and highlight deals that need attention.”

2. The assistant identifies the authorized team and reporting period.

3. It retrieves permitted opportunities, stage history, expected close dates, activities and follow-up status.

4. The system calculates pipeline totals, stage movement and inactivity using defined business rules.

5. The AI summarizes notable changes and flags deals with evidence such as no recent activity or repeated close-date changes.

6. The assistant presents the source opportunities, metric definitions and suggested next actions.

7. The manager may ask for draft follow-up tasks. The system previews the tasks and requires approval before creation.

9.7 AI Limitations and Safe Behaviour

- The assistant must not claim that an email was sent, a record was updated or a task was created until the business operation confirms success.
- The assistant must not invent customer commitments, proposal prices, dates, technical fixes or policy statements.

- Retrieved documents may contain malicious or irrelevant instructions. Document text is information to analyze, not authority to override system rules or permissions.
- The assistant must not reveal hidden prompts, credentials, another tenant's information or records outside the user's permissions.
- Bulk changes, external messages, deletions and customer commitments require explicit confirmation and audit logging.
- AI failures must not block core CRM work. Users must be able to perform key tasks manually if the AI service is unavailable.

10. Business Rules

Rule ID	Business rule
BR-001	Every lead, account, contact, opportunity and ticket belongs to exactly one tenant.
BR-002	A lead must have a minimum identity set defined by the organization (for example, name or company plus at least one contact channel) before it can be saved as active.
BR-003	Duplicate detection may suggest a match; uncertain records must be reviewed before merge.
BR-004	A lead conversion must preserve source lead details, activities and ownership history.
BR-005	An account may represent a prospect before it becomes a customer. Customer status must follow an organization-defined conversion event.
BR-006	A contact must be linked to an account or lead, except where a clearly identified standalone prospect contact is allowed.
BR-007	An opportunity must have an owner, account, stage and expected value or an explicit "value not yet known" state.
BR-008	An opportunity marked Won requires a final outcome date and final value; an opportunity marked Lost requires a loss reason.
BR-009	Opportunity stage changes must be timestamped and attributable to a user or approved workflow.
BR-010	An active opportunity should have a next action or an explicit reason why no next action is required.
BR-011	A proposal sent to a customer must be an approved, versioned proposal; a sent version must not be silently overwritten.
BR-012	Proposal commercial values and contractual statements require authorized human approval.
BR-013	A task must have an owner and due date unless the task type explicitly permits otherwise.
BR-014	Internal ticket notes must never appear in the customer portal or customer-visible email.
BR-015	A ticket must have a linked account and an identifiable requester or documented source.
BR-016	Ticket SLA calculations must use the tenant's configured business hours, holidays and priority policy.
BR-017	A ticket can be reopened only under the organization's configured reopening policy; previous resolution history remains available.
BR-018	Only published and authorized knowledge content can be retrieved for general RAG answers.
BR-019	Customer-specific documents must be retrieved only for users authorized for that customer.

BR-020	AI-generated content is a draft or recommendation until an authorized user accepts it.
BR-021	All AI tool calls must use an approved tool catalogue and the initiating user's permissions.
BR-022	External communications require an authorized sender and must be previewable before dispatch.
BR-023	AI-generated metrics must be based on defined metric calculations rather than free-form model arithmetic.
BR-024	Machine-learning scores must be displayed as estimates with model/version and evaluation status available to administrators.
BR-025	Tenant-level data must not be used to train a shared model unless the tenant has explicitly agreed under a documented data policy.
BR-026	A user deactivation must not remove historical records or audit events created by that user.
BR-027	Bulk export, bulk update and merge operations require dedicated permissions and must be audited.
BR-028	A customer portal user may access only the records explicitly shared with their customer organization and permitted contact scope.

11. Business Data Requirements and Data Quality

11.1 Core Business Data

Data group	Examples	Business quality expectation
Organization / user	Tenant profile, employees, teams, roles.	Unique identity, active status, correct role and ownership.
Lead	Name, organization, contact channel, source, enquiry, qualification.	Source and owner captured; duplicate review; status meaningful.
Account / contact	Company details, people, designations, relationship status.	One usable organization profile; contacts linked correctly.
Opportunity	Need, value, stage, owner, expected close, outcome.	Stage history and outcome reasons complete enough for analysis.
Proposal	Version, scope, commercial terms, approval, send outcome.	Version traceability and approved customer-facing copy.
Activity	Call, meeting, task, note and follow-up.	Relevant record link, owner and useful date/time.
Ticket	Requester, issue, category, priority, status, resolution.	Clear issue description, ownership, customer-safe communication.
Knowledge content	Article, document, category, version, visibility.	Owner, approved status, access classification and review date.
AI evidence	Prompt context references, retrieved sources, tool history, feedback.	Traceable, permission-aware and retained according to policy.

11.2 Data Import Requirements

- Initial migration may use CSV templates for leads, accounts, contacts and opportunities.
- Before import, the organization should map source columns to CRM fields and review mandatory-field gaps.

- The import should provide a preview, duplicate warnings, validation messages and a downloadable result summary.
- Failed rows should not cause valid rows to be lost; import outcomes must be clearly reported.
- Imported records should identify their origin as imported and retain import batch metadata for traceability.

11.3 AI Data Readiness

AI quality depends on the quality and coverage of business data. Predictive ML features should not be activated just because a model can be trained. The business must first have enough historical records with reliable outcomes, consistent definitions and representative coverage. For a new tenant with little history, CRMPro AI should use transparent rule-based suggestions or omit predictive scores rather than present misleading precision.

12. Integration requirements

Integration	Business purpose	Initial expectation
Website enquiry form	Capture leads without manual re-entry.	Should: accept approved form submissions and record lead source.
Email service	Send and receive customer communication and associate it with CRM records.	Should: support a controlled email integration; initial release may use approved outbound email only.
Calendar service	Schedule meetings and reduce conflicting manual entries.	Could: add after core activity management is stable.
File storage	Store proposal and knowledge attachments.	Must: provide reliable attachment access subject to tenant and record permissions.
Support mailbox	Create tickets from a support address.	Could: add after ticket workflow is stable.
External delivery/project tools	Link a won opportunity to delivery work.	Future: link or handoff only; not a full project management replacement.
LLM / embedding provider	Provide generative and retrieval capabilities.	Must for AI release: configurable provider boundary, data handling review and usage controls.
ML inference capability	Provide validated prediction results.	Future: only for models that pass data-readiness and evaluation gates.

13. Reporting Definitions and Success Measures

A report is only useful if its calculation is consistent. The following definitions are proposed and should be confirmed by the product owner. Filters must respect tenant and role permissions.

Metric	Proposed business definition	Important qualification
Lead conversion rate	Number of leads converted in the period divided by the number of leads eligible for conversion in the same cohort/period.	The organization must choose cohort-based or period-based reporting and use it consistently.
Opportunity win rate	Won opportunities divided by all decided opportunities (Won + Lost) in the selected cohort or period.	Open opportunities are excluded from the denominator.
Pipeline value	Sum of current estimated values for open opportunities under the selected	Do not treat as booked revenue.

	filters.	
Weighted pipeline	Sum of opportunity value multiplied by probability for open opportunities.	Probability source must be identified: user-entered, stage default or validated model.
Average sales cycle	Average elapsed time between the defined opportunity start date and Won date for won opportunities.	Start event must be consistently defined.
First response time	Elapsed business time from ticket creation to the first public human response.	Exclude automated acknowledgement unless policy explicitly counts it.
Resolution time	Elapsed business time from ticket creation to resolved timestamp.	Define pause rules for Waiting for Customer.
SLA attainment	Tickets meeting the applicable response/resolution target divided by tickets with a target due in the period.	Use configured business hours and pause rules.
AI draft acceptance rate	AI-generated drafts accepted with minor/no edits divided by drafts reviewed.	Interpret alongside user feedback and quality sampling; high acceptance alone does not prove correctness.
RAG grounded-answer rate	Sampled RAG answers whose key claims are supported by retrieved authorized sources.	Requires a reviewed evaluation set and sampling method.
Agent task success	Approved agent workflows completed the intended business operation without an unhandled failure.	Track partial completion and user correction, not just tool execution.

13.1 Business-Level Product KPIs

- Percentage of active leads with an owner and next action.
- Median time from new lead creation to first meaningful response.
- Percentage of open opportunities with a current expected close date and next action.
- Opportunity win rate and average sales cycle, with clearly defined cohort rules.
- Percentage of tickets acknowledged within the configured target.
- Average resolution time and percentage of tickets reopened.
- Percentage of AI recommendations reviewed, accepted, edited or rejected.
- RAG answer groundedness and correct source attribution on a reviewed test set.

14. Non-Functional and Operational Requirements

Area	Business requirement
Security	The product must protect tenant records, user identities, customer data, uploaded documents and AI context. Access must be role- and record-aware.
Privacy	The organization must understand what data is sent to an AI provider, for what purpose, retention expectations and available controls.
Tenant isolation	A user, search, report, background process or AI retrieval request must never cross tenant boundaries.
Availability	Core CRM workflows should remain usable when AI providers are unavailable. AI must be an optional assistance layer, not a single point of failure for record management.
Performance	Common list, detail and save operations should feel responsive under expected tenant workloads. Long-running imports and AI processing should provide status rather than freeze a user session.

Scalability	The product should accommodate growth in tenants, users, records, tickets, documents and AI requests without redesigning core business concepts.
Auditability	Important record changes, exports, approvals, AI tool actions and permission changes must be traceable.
Usability	Screens and workflows should use consistent terminology, reduce duplicate data entry and make ownership and next action visible.
Accessibility	Core workflows should support keyboard use, readable contrast, clear labels and accessible validation messages.
Retention and deletion	The organization must define retention periods for CRM records, files, email content and AI interaction logs, subject to applicable obligations.
Backup and recovery	The service must maintain recoverable copies of business data and document attachments and periodically verify recovery procedures.
Observability	The service should track errors, latency, queue failures, AI provider failures, token/cost usage and model/prompt versions.
Portability	Organizations should be able to export their core business records and attachments in a documented format under controlled permissions.

14.1 AI-Specific Risk Controls

Risk	Required control
Hallucinated answer	Ground answers in retrieved records/documents, cite evidence and abstain when unsupported.
Cross-tenant data leakage	Enforce tenant and authorization filters before retrieval and before tool execution; test isolation explicitly.
Prompt injection in documents	Treat document text as untrusted data; do not allow it to override system policy or trigger tools.
Unauthorized action	Use allow-listed tools, permission checks, previews and explicit approval for consequential actions.
Incorrect classification	Show suggestions as provisional, allow correction and route low-confidence items to manual triage.
Biased or misleading scores	Review feature choices and outcomes; monitor performance across relevant groups where lawful and appropriate; provide non-AI fallback.
Sensitive data exposure	Minimize model context, redact where suitable, restrict logs and configure provider/data processing terms.
Provider outage or rate limit	Fail gracefully; preserve drafts where possible and allow manual work to continue.
Uncontrolled cost	Apply tenant-level usage limits, monitoring and optional feature controls.

15. Business Acceptance Criteria and UAT Scenarios

The following scenarios define the minimum observable business outcomes for acceptance. Detailed test cases should include positive, negative, permission and boundary conditions.

UAT ID	Scenario	Acceptance criteria
UAT-001	Create and assign a lead	An authorized user creates a lead with required details; a manager assigns it;

		owner and assignment history are visible.
UAT-002	Detect duplicate lead	A likely duplicate is flagged; a user can compare records; no record is merged without an authorized confirmation.
UAT-003	Convert a qualified lead	A qualified lead is converted into linked account/contact/opportunity records; the original lead and activities remain accessible.
UAT-004	Track a sales opportunity	An authorized user moves an opportunity through configured stages; each transition is recorded; Won/Lost required fields are enforced.
UAT-005	Prepare and revise a proposal	A proposal is created from an opportunity, revised as a new version and approved before sending; sent version is preserved.
UAT-006	Record customer activity	A meeting or call is recorded against the correct customer/opportunity; it appears in the account timeline and creates a follow-up task.
UAT-007	Create and resolve ticket	A customer creates a ticket; support assigns, responds, resolves and closes it; internal notes are not visible to the customer.
UAT-008	Customer portal isolation	A portal user sees only records shared with their own organization and cannot access another tenant or customer's ticket.
UAT-009	RAG answer from approved article	An employee asks a question answered by a published article; answer includes the article source; draft/restricted articles are not retrieved.
UAT-010	RAG cannot find answer	An employee asks a question not supported by the knowledge base; assistant clearly states that it cannot find a reliable answer.
UAT-011	AI ticket response draft	AI uses ticket context and approved article to draft a reply; agent can edit; no email is sent until an authorized user approves.
UAT-012	CRM data question	A manager asks for open opportunities above a chosen value; assistant returns records based on actual CRM filters and current permissions.
UAT-013	AI agent prepares tasks	A manager requests follow-up tasks for stale deals; assistant shows proposed tasks and owners; no task is created until approval.
UAT-014	AI provider unavailable	Users can still create and update core CRM records; AI features show an understandable temporary-unavailable message.
UAT-015	Tenant isolation	A user from Tenant A cannot retrieve Tenant B records through screens, search, reports, exports or AI assistant.
UAT-016	Audit trace	A proposal approval, ticket reassignment, bulk export and AI action each create an attributable audit event.

15.1 Definition of Business Readiness

- Core user journeys can be completed without manual database intervention.
- Every active lead, opportunity and ticket has visible ownership and status.
- Role permissions and tenant separation pass UAT scenarios.
- AI features are clearly labeled as generated/suggested, provide evidence where applicable and have manual fallbacks.
- Support staff can work tickets without relying on AI.
- Business reports use approved definitions and reconcile to underlying records.
- A public demo environment contains synthetic sample data and does not expose private customer information.

16. Recommended Release Plan

The BRD describes the target product capability. The following staged delivery plan protects the project from becoming too large for a portfolio implementation. It prioritizes a working product that can be demonstrated publicly, then adds deeper AI capability based on data and usage readiness.

Release	Business scope	AI scope	Exit outcome
Release 1 – Core CRM MVP	Tenant setup; users/roles; dashboard; leads; accounts/contacts; opportunities; activities; basic reports.	Lead/enquiry summary and follow-up draft may be included if core flows are stable.	A sales team can manage a lead-to-opportunity process in the live application.
Release 2 – Customer Support	Ticket lifecycle; assignment; customer portal; SLA basics; knowledge base and document management.	Ticket classification suggestion, conversation summary, RAG knowledge search and suggested reply.	A customer can raise a ticket and a support agent can resolve it using knowledge assistance.
Release 3 – Agentic CRM	Customer 360, expanded reporting, proposal workflow, operational refinements.	Read-only CRM assistant, sales pipeline analysis, customer summary workflow and approval-based draft task creation.	A manager can ask business questions and run bounded multi-step workflows.
Release 4 – Predictive Intelligence	Data quality improvements, outcome reporting and model governance.	Lead scoring or deal win probability only after adequate historical data and evaluation.	Predictive scores are demonstrably useful, calibrated and monitored—or deliberately withheld if data is insufficient.

16.1 Recommended First Public Demo

For a credible public portfolio release after the customer support and knowledge-base capabilities are available, the demo should let a visitor sign in with a demo account, inspect sample IT-service-company data, create or update a lead, move an opportunity, view customer history, create a support ticket, search the knowledge base and try an AI assistant. The demo should use synthetic data, enforce safe rate limits and provide a reset mechanism so visitors can explore without damaging the shared experience.

16.2 Explicitly Defer Predictive ML if Data Is Not Ready

A new CRM has no meaningful history of its own. A model trained on a small or synthetic dataset may look impressive but can give misleading results. For the initial release, use transparent qualification checklists and rule-based risk indicators. Add a supervised model after the product has a sufficiently representative dataset with reliable labels such as converted/not converted or won/lost, and after a baseline comparison and calibration review.

17. Assumptions, Dependencies and Open Decisions

17.1 Assumptions

- The initial market is small and medium-sized B2B IT service providers, not all CRM industries.
- The primary language for the first release is English; localization may be added later.
- Tenants maintain their own users, records, configuration and approved knowledge content.
- The organization remains responsible for the correctness of customer data, proposal terms and published knowledge.
- AI providers may process selected context to generate responses; provider terms and data handling must be reviewed before production use.
- Customers access the portal only after an internal administrator invites or approves them.
- The customer portal is limited to ticket and approved shared information in the initial release.

17.2 Dependencies

- Product owner approval of sales stages, ticket statuses, role templates and metric definitions.
- Availability of sample or migrated CRM records for demonstrations and testing.
- Approved support documents and knowledge articles for RAG testing.
- Email service configuration if outbound communication is included.
- Privacy and security review of AI provider settings, tenant data handling and retention.
- Named business users who can review end-to-end workflows and AI outputs.

17.3 Open Business Decisions

Decision	Recommended baseline	Must be confirmed by
Customer portal scope	Ticket creation, ticket view/reply and published knowledge access only.	Product owner
Proposal approval	Internal approval required above a configurable value or for selected service types.	Business owner
Account conversion	Set Customer after proposal acceptance/authorized commercial confirmation.	Sales owner
Ticket SLA policy	Business hours and priority-based targets configurable per tenant.	Support manager
AI data retention	Store minimal context and audit metadata under configurable retention; do not use tenant data for shared model training by default.	Organization owner / privacy reviewer
AI-generated outbound messages	Draft only; human approval before sending.	Business owner
ML activation threshold	Define minimum data-volume and model-quality gates before launch.	Product owner / ML owner
Multi-language support	English in Release 1; assess demand	Product owner

	before localization.	

18. Business Risks and Mitigations

Risk	Potential impact	Mitigation
CRM becomes too broad	Long development cycle and unfinished modules.	Protect MVP scope; exclude ERP, project delivery, accounting and marketing automation.
Poor CRM data quality	Unreliable dashboards, summaries and AI suggestions.	Require key fields, duplicate review, data completeness indicators and ownership.
Users do not maintain records	Customer history and pipeline become stale.	Make updates low-effort; show next actions; define process ownership and management review.
AI answers appear confident but are wrong	Loss of trust or incorrect customer commitments.	Ground answers, cite evidence, abstain, evaluate and require human review.
RAG retrieves restricted information	Confidential customer or internal information leaks.	Apply permission filters before retrieval; test tenant and document isolation.
Insufficient data for predictive ML	Scores create false confidence.	Defer ML until representative outcomes and validation are available.
Excessive AI usage cost	Operating expense becomes unpredictable.	Usage budgets, rate limits, model routing and per-tenant monitoring.
Email integration creates duplicate or misfiled messages	Confusing communication history.	Explicit linking rules, sender verification and manual reassignment.
Customer portal exposes internal notes	Confidential internal discussion reaches customers.	Separate public replies and internal notes at the business data and permission level.
AI automation changes records unexpectedly	Operational errors and loss of user confidence.	Preview, approval, idempotency, audit trail and limited tool permissions.

19. Glossary

Term	Meaning
Account	An organization with which the IT company has a prospect or customer relationship.
Activity	A logged interaction or work item such as a call, meeting, task or email.
Agent	An AI-driven workflow that can plan bounded steps and use approved tools to complete a defined task.
AI grounding	Using approved records or documents as evidence for an AI response.
Contact	A person associated with an account or lead.
Customer 360	A consolidated view of the customer's relationship, people, sales, interactions and support history.
Embedding	A numerical representation of text used to find semantically related content.
Lead	A potential buyer or business enquiry that has not yet become a qualified opportunity/customer.
LLM	Large Language Model; a model that understands and generates natural language.
Opportunity / Deal	A specific potential commercial sale tracked through sales

	stages.
RAG	Retrieval-Augmented Generation; retrieve relevant approved content and provide it to an LLM to generate a grounded answer.
SLA	Service Level Agreement or internal service target defining expected response/resolution times.
Tenant	An independent organization using the SaaS product.
Tool calling	A model requesting a defined application operation, such as retrieving a ticket or preparing a task.
Win rate	The percentage of decided opportunities that are won, using an agreed reporting definition.

20. Product Owner Recommendation and Baseline Scope

The recommended product baseline is a focused IT-services CRM with a complete lead-to-customer relationship flow and a practical support workflow. The first product should not attempt to reproduce Salesforce, HubSpot or a full helpdesk suite. It should provide a coherent, usable set of workflows that can be understood by a visitor and demonstrated end to end.

Recommended first-release scope
Tenant/user administration; lead management; account/contact management; opportunity pipeline; activities and follow-ups; dashboard and basic reports. Add ticketing, customer portal and RAG knowledge assistance as the next major release. Add bounded AI agents after the underlying business workflows and permissions are reliable. Add predictive ML only when real outcome data supports it.

20.1 What Makes This a Real Product Rather Than a Demo

- Records are connected through a real lifecycle, not isolated CRUD screens.
- Every important object has ownership, status, history and access rules.
- The customer portal provides an actual external user journey.
- Support knowledge is governed and can be searched with cited evidence.
- AI performs useful work inside the CRM—summarizing, retrieving, recommending and preparing actions—not just chatting.
- AI actions are permission-aware, reviewable and audited.
- The public demo uses coherent synthetic business data and can be safely reset.
- The BRD, user guide, architecture overview, evaluation report and demo walkthrough explain how the product works.

20.2 Approval Checklist

Review item	**Decision / owner**
Target market and company size assumption	To confirm – Product owner
Lead, account, contact and opportunity definitions	To confirm – Sales representative
Sales stages and conversion rules	To confirm – Sales manager
Ticket statuses, priorities and SLA model	To confirm – Support manager
Customer portal visibility and user onboarding	To confirm – Product owner
AI data sources, permissions and retention	To confirm – Organization owner / privacy reviewer
Human approval requirements for AI actions	To confirm – Business owner
MVP release boundary and deferred capabilities	To confirm – Product owner

End of CRMPro AI Business Requirements Document